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Undergraduate Research on Elizabeth River Simulation

Introduction:

This project focuses on simulating flooding in the Elizabeth River using Blender and Unity. The goal is to recreate how water behaves during real flood events and visualize it in a 3D and VR environment. By combining real geographic data with fluid simulation, this project helps show how flooding spreads across terrain and which areas are most at risk.

To make the simulation accurate, the project uses **metrics**, which are measurable values used to compare results. In this case, metrics include things like **water level height (for example, 5 ft from Hurricane Sandy), how fast water spreads, and which areas get flooded first**. These measurements are important because they allow us to check if the simulation matches real-world data. If the same areas flood in both the simulation and historical events, the model is considered reliable.

Github repository for files: https://github.com/fabianafazioo/Elizabeth-River-Water-Simulation.git

PART 1 : BLENDER

Step 1 — Add files

1. Download IMG file from this link: https://www.sciencebase.gov/catalog/item/581d2b84e4b08da350d63f4c
2. Open terminal and run theses commands:

   brew install gdal

   gdalinfo --version

   gdalinfo your_file.img | head -n 40

   gdal_translate -of GTiff -co COMPRESS=DEFLATE -co TILED=YES \ your_file.img elizabeth_dem.tif

   gdalwarp -t_srs EPSG:26918 -r bilinear -co COMPRESS=DEFLATE -co TILED=YES \ elizabeth_dem.tif elizabeth_dem_utm18.tif

```
gdal_translate -projwin 566133.05 4501851.18 567133.05 4500851.18 \ elizabeth_dem_utm18.tif elizabeth_clip.tif

gdalinfo -mm elizabeth_clip.tif | grep -E "Computed Min/Max"

gdal_translate elizabeth_clip.tif height.png \ -ot UInt16 -of PNG \ -scale ZMIN ZMAX 0 65535
```

Step 2 — Use NJFloodMapper for Metrics

1. Go to https://www.njfloodmapper.org/

Select:

- Total Water Levels Tool
- Click on Flood Event
- Click on Historical Events → Select Hurricane Sandy [10/29/2012] → Select 5 ft water level
- Click Add
- Now analyze the areas that are getting flooded faster.

At the end of this project, go back to this simulation and analyze your results from blender or Unity to ensure that you have an accurate model where the same areas are getting flooded as Hurricane Sandy.

Step 3 — Blender

1. Create a new project
2. Press Shift + A
3. Select the plane → name: Terrain
4. Go into Edit Mode
    a. Press **A**
    b. Right-click > Subdivide: 8 times
5. Return to Object Mode
    a. Select the plane
    b. Go to modifiers tab
    c. Add modifier > displace
    d. Click new (next to texture)
    e. Click the texture properties tab (checkerboard icon)
    f. Under image > click open > navigate to your_file.tif > open image
    g. Go to Image Texture Settings > color space > sRGB → Non-Color

6. Go into Edit mode again > Subdivide : 4 times (higher resolution)

- Click on Terrain
- Go to **Physics Properties**
- Enable **Fluid**
- Set:
    - Type: **Effector**
    - Effector Type: **Collision**
    - **Sampling substeps: 3**
    - **Surface thickness: 0**
    - **Enable → Use Effector**
    - **Enable → is planar**

Step 4 — Add Water Domain

1. Shift + A → Mesh → Cube
2. Scale it so it **covers exactly the scene**
3. With the cube selected:
    - Physics Properties → **Fluid**
    - Type: **Domain**
    - Domain Type: **Liquid**

Domain settings (IMPORTANT)

- Resolution Divisions: **128**
- Time Scale: **1.0**
- **Turn on: adaptive time steps**
- **CDL number: 2**
- **Timesteps maximum: 8**
- **Minimum: 2**
- Turn off all border collisions except “bottom”
- Enable liquid → Select simulation method → **FLIP**
    - **Enable → fractional obstacles**
- Enable **Mesh**
    - **Upres factor: 4**
    - **Particle radius: 2**

Step 5 — Prepare and Refine River Geometry

The terrain imported from GIS data already contains the general shape of the Elizabeth River. However, the river channel must be refined so it can properly hold water and act as the starting point for the flood simulation.

Steps

- Enter **Sculpt Mode** on the terrain mesh.
- Use the **Smooth Brush first** to remove irregular terrain artifacts from the imported GIS mesh.
- After smoothing, switch to the **Draw Brush** to carve the river channel deeper.
- Increase the **depth of the river bed** so water can accumulate and flow naturally.

Edge Refinement

- Use a **smaller sculpting brush** to shape the river edges and banks.
- This helps maintain realistic shorelines without affecting surrounding terrain.

Important Terrain Adjustment

- Be careful around the **1-9 highway area**.
- Avoid damaging terrain that represents road infrastructure.
- Instead, focus on **deepening the river channel in this section**.

Sculpting Workflow

Repeat the following process until the riverbed looks continuous and realistic:

1. **Carve the river**
2. **Smooth the terrain**
3. **Redraw sections if needed**

The goal is to create a **clear and deep river path** that will serve as the water source for the flood simulation.

Step 6 — Add Water Source (Spheres)

1. Shift + A → Mesh → **Spheres**
2. Add **spheres along the river channel**.
3. These spheres will act as **water emitters**.
4. Space them evenly along the river.

Fill Empty Areas

- If you see **gaps between spheres**, add additional spheres.
- The river must be **fully populated with fluid sources** so water flows continuously.

Configure Sphere Physics

For each sphere:

- Go to **Physics Properties**
- Select **Fluid**
- Set the object type to **Flow**

Configure the following:

- **Flow Type:** Liquid
- **Flow Behavior:** Geometry

This allows the spheres to emit water particles into the simulation.

Step 7 — Generate The Initial River Water State

The goal is to let the river establish water flow before converting it into a flooding source.

Steps

1. Run the simulation until approximately **Frame 33**.
2. At this point, the river should contain flowing water.

Duplicate the Water Domain

- Duplicate the fluid domain.
- Convert the water simulation to a **mesh**.

Reset the Timeline

- Rewind the animation back to **Frame 0**.

Place the Water Mesh

- Move the generated **water mesh back into the river channel**.

Step 8 — Convert The River Mesh into a Flood Source

Once the water mesh is placed in the river:

- Select the **mesh object**
- Go to **Physics Properties**
- Set it to **Fluid**

Configure the mesh as:

- **Type:** Flow
- **Flow Type:** Liquid
- **Flow Behavior:** Geometry

This allows the existing river water to act as the **starting flood source**.

Re-bake the Simulation

- Return to the **Fluid Domain settings**
- Bake the simulation again

Now the river water will begin **spreading outward and flooding the terrain**.

Step 9 — More Details to Add

During simulation, several terrain adjustments may be necessary.

If the River Overflows Too Early

- Enter **Sculpt Mode**
- Use the **Smooth Brush**

Shortcut:

CTRL + R

Repeat smoothing and rebaking until the river behaves correctly.

If the Terrain Has Sharp Points or Irregular Geometry

1. Switch to **Edit Mode**
2. Highlight the problematic area
3. Adjust vertices using:

G → Y

G → Z

This allows you to move geometry along the terrain.

Final Terrain Cleanup

After editing geometry:

- Return to **Sculpt Mode**
- Apply the **Smooth Brush**

Important reminder:

- **Remesh frequently**
- **Save your project often**

Remeshing ensures the terrain maintains even polygon distribution and avoids simulation artifacts.

Step 10 — Apply Water Material

1) Assign material to the LIQUID MESH

- ❌ Not the Domain cube
- ❌ Not the Flow object
- ✅ **Select the Liquid mesh** generated by Mantaflow

Then:

- Material Properties → **New**
- Name it: Water_Material

Then:

- Principled BSDF
- Base color HEX: #84CEE751
- Roughness: 0.050
- IOR: 1.5
- Alpha: 0.205

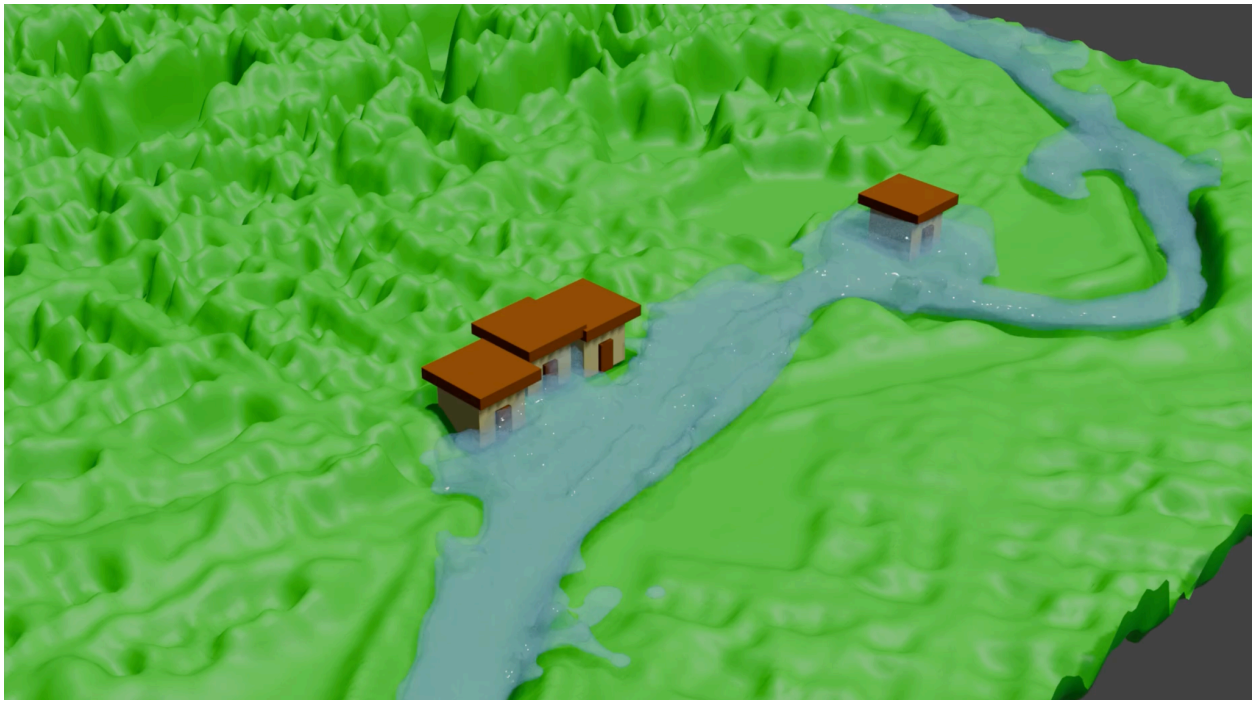

Figure 1: Blender Scene

PART 2: Unity

Blender → Unity VR Flood Visualization Pipeline

This section explains how to export the terrain and water simulation from Blender and correctly import them into Unity so the scene can be used for VR visualization.

Step 1 — Prepare the Blender Scene for Export

Before exporting anything, the Blender scene must be cleaned so Unity can read the geometry correctly.

Convert Fluid Simulation to Mesh

Unity cannot read Blender fluid simulations directly. The simulation must be converted into geometry. Inside Blender do this:

Steps:

1. Select the **Fluid Domain Object**.
2. Go to the **Physics Properties panel** (right side of the screen).
3. Scroll to the **Fluid section**.
4. Under **Cache**, ensure the simulation has already been **baked**.
5. Under **Mesh**, enable the **Mesh checkbox**
6. Click **Bake Mesh**

Blender will generate a mesh surface representing the water.

Apply Modifiers

For every object being exported:

1. Select the object.
2. Go to the **Modifier Properties panel** (wrench icon).
3. If modifiers exist (Subdivision, Remesh, etc.), click **Apply**.

This converts procedural changes into permanent geometry that Unity can read.

Clean the Scene

Remove objects that should not be exported. Recommended export objects:

- Terrain mesh
- River mesh
- Water mesh

Step 2 — Export the Blender Scene

Unity works best with **FBX files**, so export the scene in FBX format.

Export the terrain ONLY if you want to run the water math in Unity.

Export initial condition of water by itself. You can do this by creating a new collection where you put the water condition and export this.

Export Steps

1. Go to the top menu:

File
→ Export

→ Click Active Collection
→ Enable Mesh
→ FBX (.fbx)

Export the File

1. Choose a folder inside your Unity project:

Example:

UnityProject/Assets/Models/

2. Name the file:

ElizabethRiverFlood.fbx

3. Click:

Export FBX

Step 3 —Import the Scene into Unity

Now open Unity and import the exported Blender scene.

Open the Unity Project

Launch Unity Hub and open your project.

Import the FBX File

In Unity:

1. Go to the **Project Window**.
2. Navigate to:

Assets

3. Drag the **FBX file** directly into the Assets folder.

OR

Use the menu:

Assets
→ Import New Asset

Select the **FBX file** and click **Import**.

Go to import package > Windows > Package Manager > Import from Tarbell > find files and import them.

Step 4 — Set Up the Terrain and Water in Unity

After importing the FBX file, you must place the scene inside the Unity environment.

Add the Terrain

1. Drag the **ElizabethRiverFlood FBX** from the **Project window** into the **Hierarchy window**.

Adjust Scale if Needed

Select the imported model in the **Hierarchy**.

In the **Inspector panel**:

Transform
→ Scale

Ensure values are:

X = 1
Y = 1
Z = 1

Step 5 — Add Lighting

Good lighting is necessary for visualizing flood conditions.

Add Directional Light

In the menu:

GameObject
→ Light
→ Directional Light

Adjust the angle to simulate sunlight.

Step 6 — Enable VR Support

If the scene will be viewed in VR (Meta Quest, etc.), configure XR settings.

Install XR Plugin

Go to:

Edit
→ Project Settings
→ XR Plugin Management

Enable the plugin for your headset.

Example:

✔ OpenXR

Add XR Camera

Delete the default camera.

Add:

GameObject
→ XR
→ XR Origin

This creates a VR camera rig.

Step 7 — Test the Simulation

Press:

Play

Inside Unity.